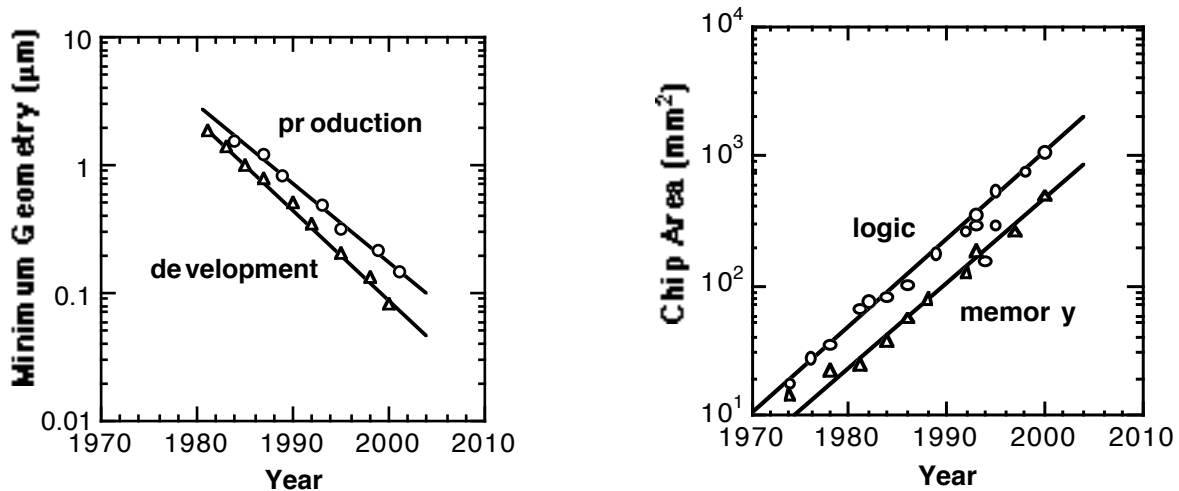


Scaling of Interconnections

Scaling of Minimum Feature size and Chip Area



Ref. A. Loke, PhD Thesis, Stanford Univ. 1999

Once the active devices and regions are fabricated they must be electrically connected to each other to make circuits. They must also be connected to the outside world through their inputs and outputs on bonding pads. Making these connections is the job of contacts, vias and interconnects. Separating the interconnects from each other is the job of dielectric layers. All of these components are part of the “metallization” or “backend” structure. The figure below is a schematic diagram showing these components in a typical integrated circuit structure. In recent times the relative importance of the backend structures has greatly increased, and will likely to continue as integrated circuit technology progresses.

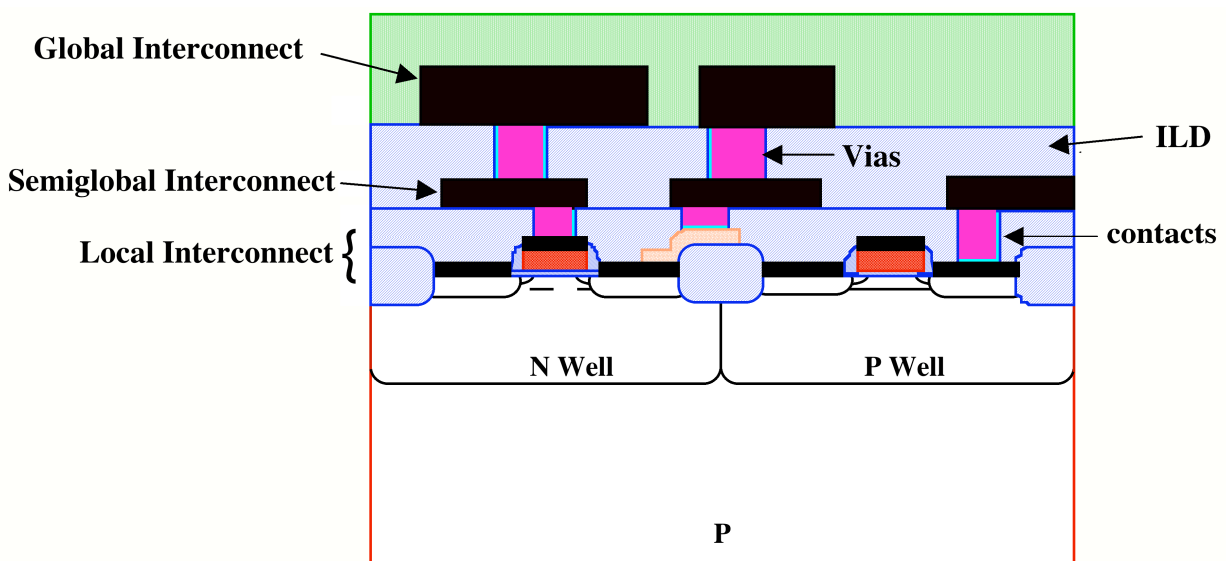


Fig. 2 Schematic cross-section of backend structure, showing interconnects, contacts and vias, separated by dielectric layers.

Interconnects can either be global or local. In general, local interconnects are the first, or lowest, level of interconnects. They usually connect gates, sources and drains in MOS technology, and emitters, bases, and collectors in bipolar technology. In MOS technology a local interconnect, polycrystalline silicon, also serves as the gate electrode material. Silicided gates and silicided source/drain regions also act as local interconnects. In addition, TiN, a by-product of a silicided gate process, can be used as a local interconnect, and W is sometimes used as well. Local interconnects can afford to have higher resistivities than global interconnects since they do not travel very long distances. But they must also be able to withstand higher processing temperatures. Global interconnects, mostly made of Al, are generally all of the interconnect levels above the local interconnect level. They often travel over large distances, between different devices and different parts of the circuit, and therefore are always low resistant metals. Local and global interconnects are shown in the multilevel schematic diagram in Figure 2 and 3.

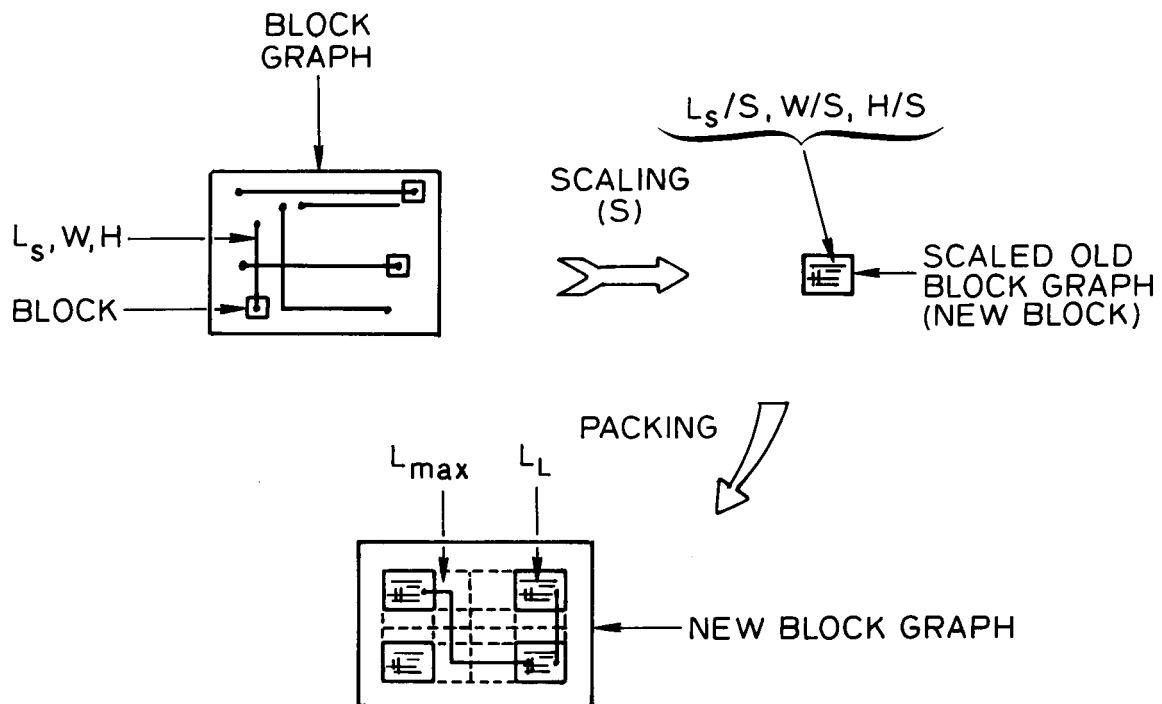


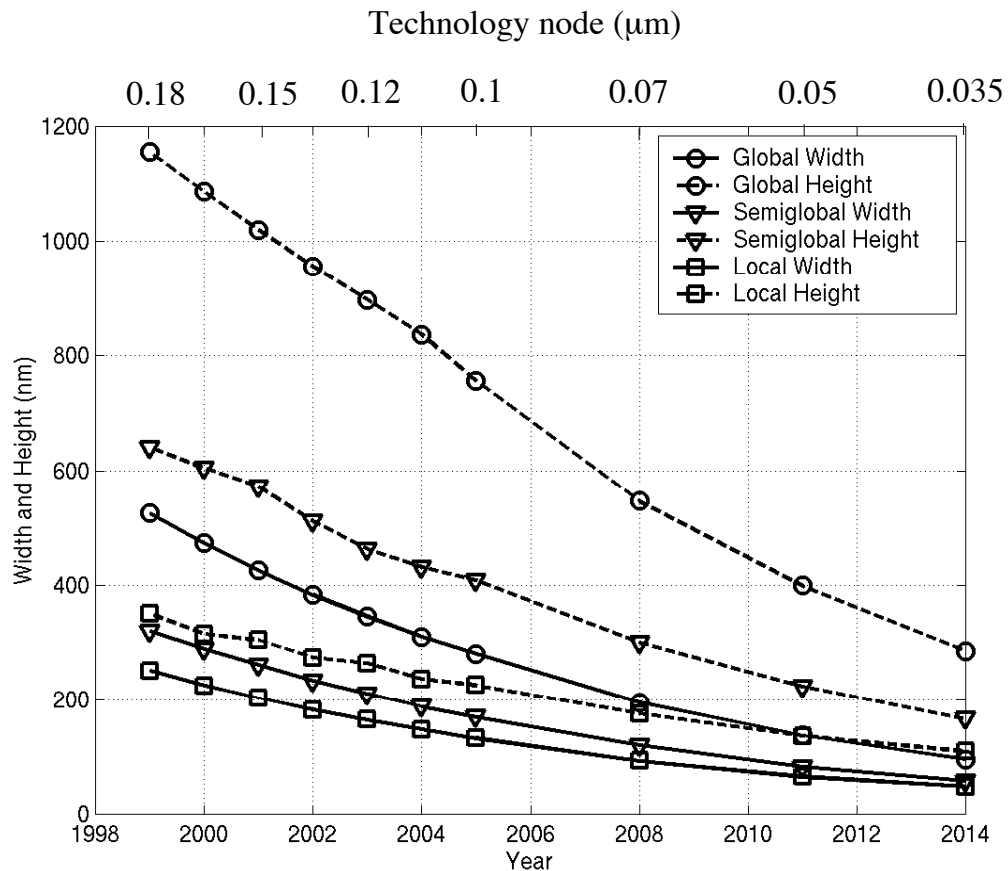
Fig. 3 Scaling of a chip and interconnections

In modern ICs as the complexity of interconnects is increasing an additional level, semiglobal interconnects between local and global has been introduced. The hierarchy of interconnects is as follows:

- Local interconnects – used for very short interconnects at the device level

- Semiglobal interconnects – Used to connect devices within a block
- Global interconnects – Used to connect long interconnects between the blocks, including power, ground and clocks.

As the device dimensions are scaled down so are that of the interconnects. The scaling trends of the height and width of the interconnects are shown below.



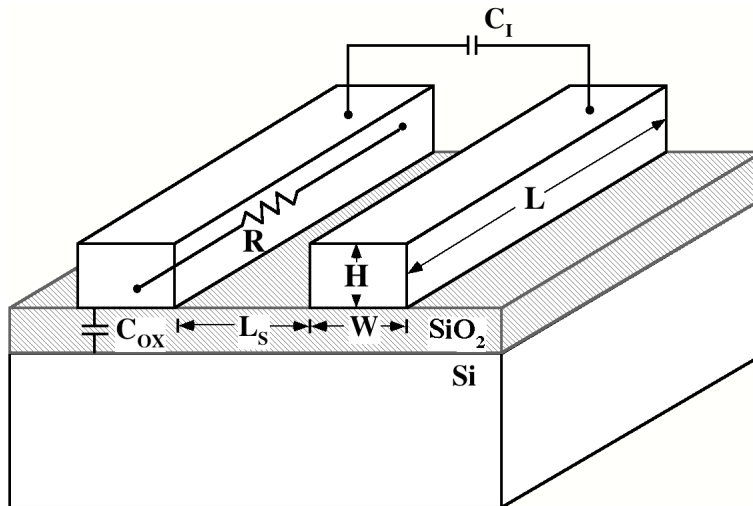
ITRS '99 dictated dimensions for local, semiglobal and global interconnects

Ohmic contacts connect an interconnect with active regions or devices in the silicon substrate. A high resistivity dielectric layer, usually silicon dioxide, separates the active regions from the first level global interconnect, and electrical contact is made between the interconnect and the active regions in the silicon through openings in that dielectric layer. At the same time in the processing, contact can be made between the first level global interconnect and the local interconnects, since they are also separated by that same dielectric layer. This is shown in Figure 2, with contacts from the lowest level global interconnect to: 1. a silicided source region in the silicon: 2. a TiN local interconnect: and 3. a silicided gate. Connections between two levels of global interconnects are usually given a different name: vias. They are made through openings in the different levels of the intermetal dielectrics. These are also illustrated in the figure. Whether they are contacts or vias, the goal of them is to electrically connect

different levels of conducting layers, and to contribute as little as possible to the total electrical resistance.

Separating interconnects from each other and from the active areas and devices are dielectric materials. Those dielectric layers separating one global interconnect level from another are called intermetal dielectrics, or IMD. (Some call these interlevel dielectrics, or ILD.) Vias connect interconnects through these layers. The layer separating global interconnects from both the substrate and local interconnects, through which the contacts are made, are given a variety of names: first level dielectric, dielectric-1, poly-gate dielectric, pre-metal dielectric, etc. - anything but intermetal dielectric. We will usually use the term “first level dielectric”. Examples of both kinds - intermetal dielectrics and first level dielectrics - are shown in Figure 2.

There are several reasons why metallization structures are getting much more attention than before. One is because of their increased importance as circuit elements decrease in size and circuit areas increase in size. Straightforward electrical analysis, following that of Saraswat and Mohammadi [ref. 1] and based on the simple structure in Figure 4, can demonstrate this. The line resistance of one of the interconnects is given by:



$$R = \rho \frac{L}{WH}$$

$$C_{ox} = K_{ox}\epsilon_o \frac{WL}{X_{ox}}$$

$$C_I = K_{ox}\epsilon_o \frac{HL}{L_s}$$

Fig 4. Interconnect structure for RC analysis. The two lines on top are the metal interconnects of dimensions W, L, and H, sitting on a SiO₂ layer. There is also SiO₂ between the metal lines.

$$R = \rho \frac{L}{WH} \quad (1)$$

where ρ is the interconnect's resistivity, and L, W, and H are the interconnect's length, width and height, respectively. The line-to-substrate capacitance is:

$$C_{ox} = K_{ox} \epsilon_0 \frac{WL}{X_{ox}} \quad (2)$$

where X_{ox} and K_{ox} are the oxide thickness and dielectric constant, respectively, and ϵ_0 is the permittivity of free space. The capacitance between the two interconnect lines is:

$$C_I = K_{ox} \epsilon_0 \frac{HL}{L_S} \quad (3)$$

where L_S is the distance between the two lines. For this simple structure, the total line capacitance can be approximated [2] by:

$$C = K_I (C_{ox} + C_I) \quad (4)$$

where K_I takes into account the fringing fields to the substrate and is approximately equal to 2. (In a multi-level interconnect structure a factor of 2 is also used for the total capacitance above and below and on both sides of an interconnect line.) The time delay of this structure can be calculated by treating it as a distributed-parameter transmission line. It can be shown that for such a system, the time delay, τ_L , is approximately equal to $0.89 RC$. Substituting in Equations 1-4 leads to:

$$\tau_L = 0.89 \cdot K_I K_{ox} \epsilon_0 \rho \frac{L^2}{WH} \left(\frac{W}{X_{ox}} + \frac{H}{L_S} \right) \quad (5)$$

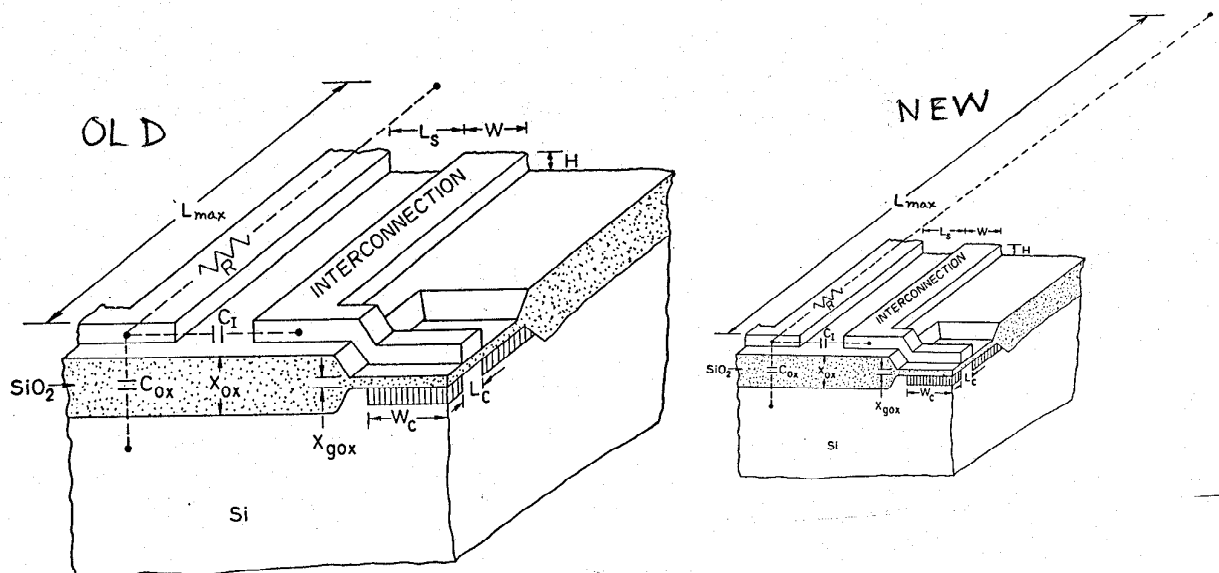


Fig. 5. Scaling of global (long distance) interconnections

Now we must consider what happens as the technology progresses in regards to the dimensions of the structure. First, L_S and W are usually close to the smallest possible dimensions possible, dictated by the lithography and etching capabilities, and designated by λ . As the technology progresses, λ gets smaller. Currently λ is 0.25 to 0.35 microns. Both X_{ox} and H shrink as λ shrinks in this ideal scaling scheme, keeping the aspect ratio, H/W , constant. It is assumed in our analysis that X_{ox} and H are equal to λ . (In the original analysis by Saraswat and Mohammadi, it was assumed that X_{ox} and H equaled 0.35λ and 0.25λ , respectively, but as we will discuss, the thicknesses have not decreased as quickly in recent years, especially for the global interconnects. We will still assume ideal, linear scaling here, but with a higher prefactor value of one.) Plugging these into Equation 5, and with K_I equaling 2, Equation 5 becomes:

$$\tau_L = 3.56 \cdot K_{ox} \epsilon_o \rho \frac{L^2}{\lambda^2} \quad (6)$$

We have said that λ gets smaller as the technology progresses. But what about L , the interconnect length? For local interconnects, L usually also shrinks as λ shrinks. Therefore, the net result is that the time delay for local interconnects stays approximately constant according to this scaling scheme. However, for global interconnects, it is a different story. As the technology progresses, the length of the global interconnects usually increases rather than shrinks as shown in Fig. 5. This is because the chip area of each new technology generation usually keeps increasing as shown in Fig. 1, forcing the global interconnects to increase in length to connect up all the areas of devices. The average length of the longest global interconnect in a circuit can be approximated by [ref. 1]:

$$L_{max} = \frac{\sqrt{A}}{2} \quad (7)$$

where A is the chip area. Equation 7 can be used to calculate the scaling factor S_c with the use of the data in Fig. 1. Inserting this into Equation 6 for global interconnects we get:

$$\tau_L = 0.89 \cdot K_{ox} \epsilon_o \rho \frac{A}{\lambda^2} \quad (8)$$

. As technology progresses, λ shrinks and A usually gets larger. By Equation 8 both of these lead to an increase in the time delay in global interconnects. Even if A stays constant, the time delay will increase due to λ shrinking. Shown in Figure 6 is Equation 8 plotted as a function of chip area for different values of λ and different interconnect materials, updated from the original figure in Ref. 1 to use the new expressions for X_{ox} and H and to include the $\lambda = 0.25$ micron graph. The interconnect delay is based on the longest global interconnect using Equation 7. Also plotted is a typical gate delay, τ_G , for that minimum feature size, which is dependent only upon λ

and not A and thus decreases as the minimum feature size decreases. One can see that for large λ 's, the gate delay dominates over the AI global interconnect delay. (And one can also see why one would not want to use polysilicon as a global interconnect.) But for smaller critical dimensions and larger chip areas, the global interconnect delay can have a large impact on the circuit performance.

Table 1. Scaling of local interconnects

Interconnection Parameter	Scaling Factor* (S)
Interconnection dimensions L_L, H, W, L_S, X_{ox}	1/S
Line resistance $R = \rho L_L / WH$	S
Line capacitance $C_{ox} = K_{ox} \epsilon_0 L_S W / X_F$	1/S
Interelectrode capacitance $C_I = K_{ox} \epsilon_0 L H / L_S$	1/S
Line response time $0.89 RC$	1
Line voltage drop IR	1
Line current density	S

*Scaling factor $S > 1$.

Table 2. Scaling of global (long distance) interconnects

Interconnection Parameter	Scaling Factor* (S)
Long-distance interconnection dimensions H, W, L_s, X_F	$1/S$
Long-distance interconnection dimension $L_{max}^{\dagger}$	S_c
Line resistance $R = \rho L_{max}/WH$	$S^2 S_c$
Line capacitance $C_{ox} = K_{ox} \epsilon_0 L_{max} W/X_{ox}$	S_c
Interelectrode capacitance $C_I = K_{ox} \epsilon_0 L_{max} H/L_s$	S_c
Line response time $0.89 RC$	$S^2 S_c^2$
Line voltage drop IR	SS_c
Line current density	S

Today, λ is commonly 0.25 to 0.35 microns, chip areas are typically 200-350 mm² (corresponding to a maximum global interconnect length of 7-9 mm) and gate delays as measured in ring oscillators are generally about 30-70 picoseconds. The global interconnect delay by this analysis is about 2-4 nanoseconds, or about 60 times that of the gate delay. This does not mean that the total interconnect delay of a circuit will be 60 times that of the total gate delay. Remember, the interconnect delay we calculated is for the longest line in the circuit. The specific circuit layout and the distribution of line lengths will determine the net interconnect delay of the circuit. What is important are the trends that these and other analyses predict: while the gate delays will keep getting smaller, the local interconnect delays will stay about the same, and the global interconnect delays will get larger and have a larger impact on the circuit performance.

Each new technology generation represents a 0.7x reduction in feature size (going from 0.5 to 0.35 to 0.25 to 0.18 microns, for example). By Bohr's results, as well as Equation 6 above, this corresponds to a 2x increase in interconnect delay per generation for a given line length. As noted by Bohr and mentioned above, the metal line aspect ratio is not typically kept constant but is increased with each technology generation largely due to the fact that the metal line thickness has stayed rather

constant in recent times rather than shrinking, as mentioned above. (The aspect ratio for Intel has increased from about 0.4 to 1.4 over the last several generations.) This means that the interconnect delay does not increase quite as fast, and has lead to an actual increase in interconnect delay per generation of about 1.3x for recent Intel logic technologies - still a significant increase. While the interconnect delay is increasing with each generation, Bohr found that the gate delay in current technologies is reduced by about 0.7x per generation, corresponding to the 0.7x decrease in feature size. As a result, the delay due to the interconnects is becoming a larger percentage of the total circuit delay time. Bohr [ref. 5] and others [ref. 4] estimate that in today's CMOS technology the delay due to the interconnects is currently about 33% of the total circuit delay or clock cycle, up from about 15-20% just a year or two previously. In the next technology generations, the interconnect delay could be more than 50% of the total circuit delay.

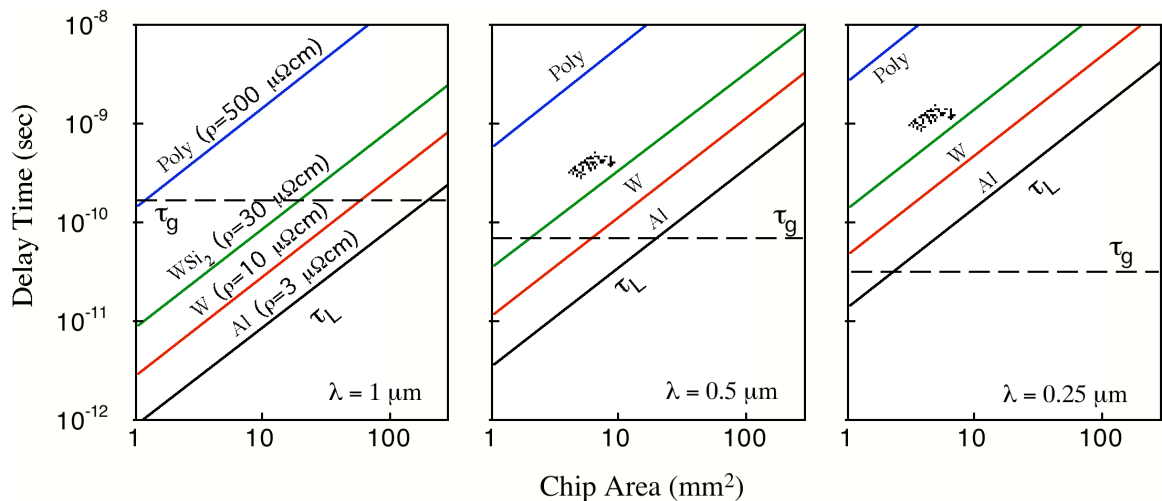
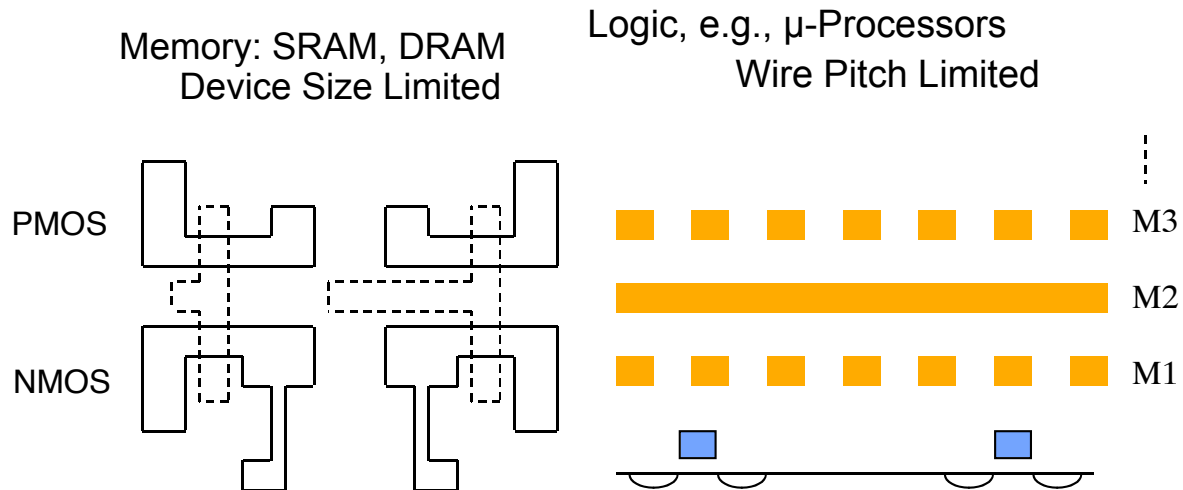


Fig. 6. Global interconnect and gate time delay versus chip area and minimum feature sizes for various interconnect materials, based on the analysis and scaling presented in the text. Interconnect delay corresponds to longest global interconnect in circuit. As the technology progresses, the interconnect delay becomes more important relative to the gate delay of a circuit. [after ref. 1]

Chip Size



Memories have very regular and dense structure. They don't need too many interconnects. As a result the device packing density is very high. Logic chips are more irregular and are dominated by communication requirements. Device packing density is *not* very high. These chips generally have larger number of interconnects and thus need more levels of them.

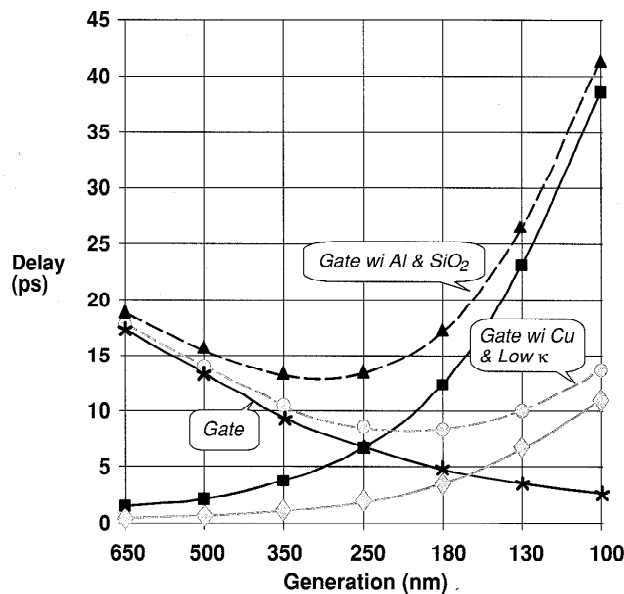


Fig. 7 Improvement in global interconnect time delay versus technology node for advanced materials (Cu and low k dielectrics)

The analysis so far has assumed that all interconnections have same area of cross section and follow same scaling rules. In a multilevel scheme that may not be true. Different layers may be optimized for different requirements. For example, for longer interconnections we may use bigger area of cross section and lower resistivity material to reduce the resistance. Smaller length interconnections could be placed in a layer of smaller area of cross section and higher resistivity material if that material has other desirable properties.

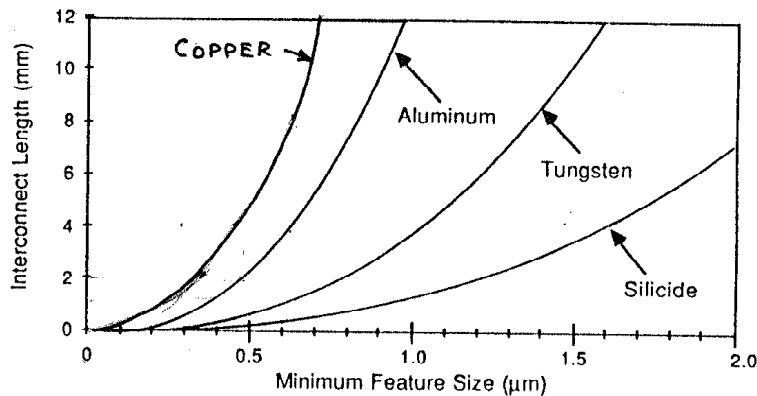


Fig. 8. Scaling long-distance interconnections using four commonly used materials resistivity materials in VLSI. The metal line width used is 1.5 times the minimum feature size. (Ref. Gardner, Meindl and Saraswat, IEEE Trans Electron Dev., Vol. ED-34, No. 3, pp. 633–643, March 1987.)

- Maximum length limited by $\tau_G = \tau_I$
- Longer wires must be fabricated with higher conductivity material – copper
- Shorter (local) wires can be fabricated with lower conductivity material – silicides, tungsten

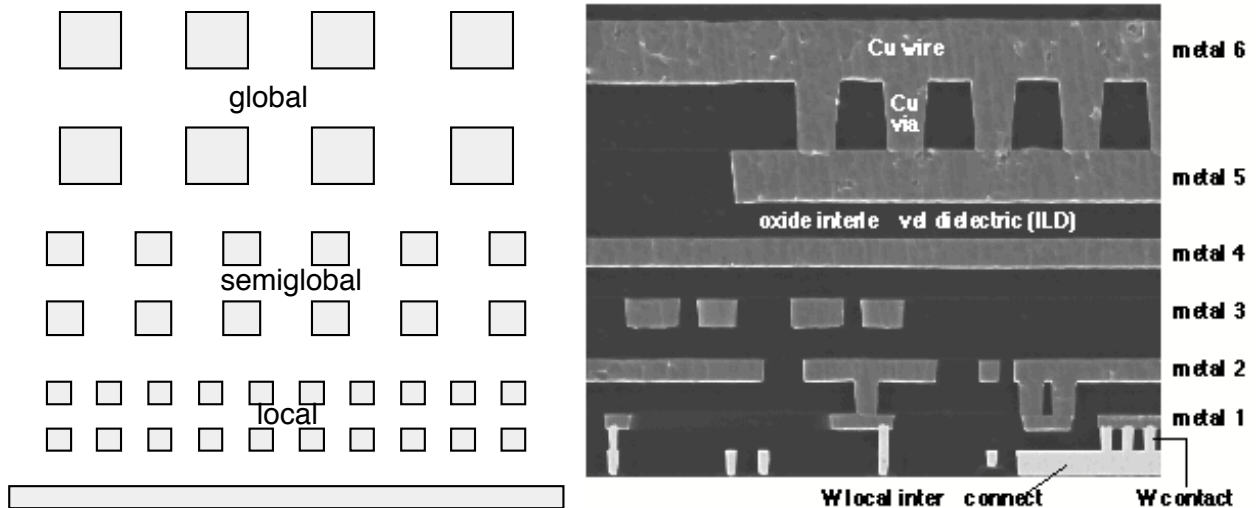


Fig. 9. (Left) A 3 tier interconnect architecture , (right) SEM of interconnect architecture with six levels of Cu wires/vias demonstrations by IBM.

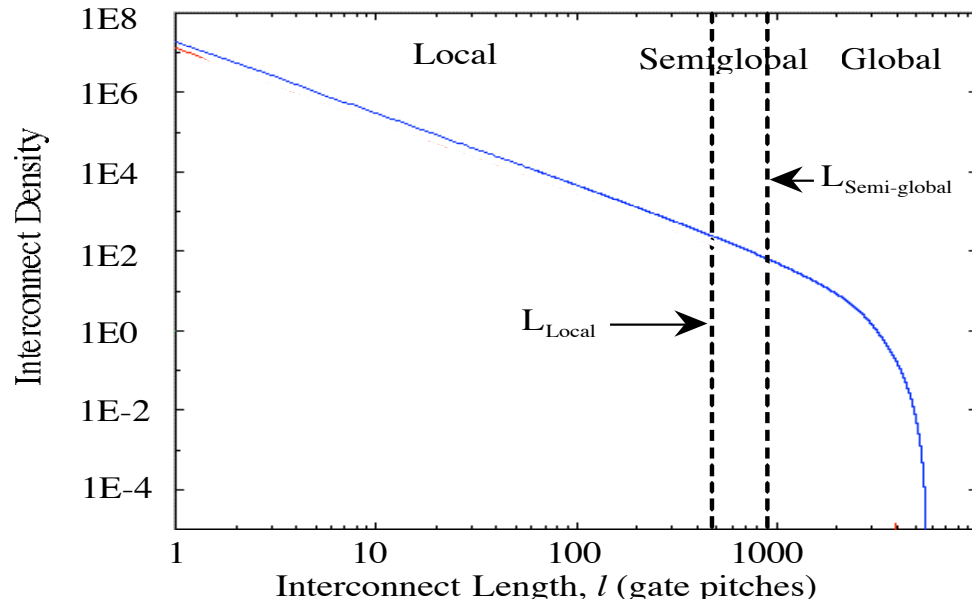


Fig. 10. Wire-length distributions (in terms of gate pitches) for a logic circuit with 180 million gates. Metal tiers determined by L_{Loc} and $L_{\text{Semi-global}}$ boundaries defined by maximum allowable delay.

- Longer wires for, clocks, power, ground and long distance communication can be placed in the global level.
- Mid length wires for communication within a block are placed in the semiglobal level.
- Very short wires where delay is primarily device governed are placed in the local level.

Use of Rent's Rule for Wire-length distributions calculations

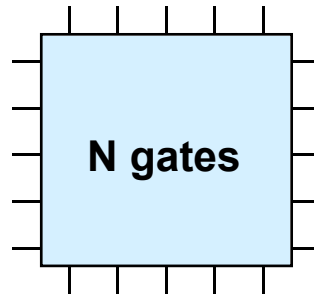


Fig. 11

Rent's Rule correlates the number of signal input and output (I/O) pins T , to the number of gates N , in a random logic network and is given by the following expression:

$$T = k N^P \quad (12)$$

where

k = average I/O's per gate

P = Rent's exponent. It denote the degree of wiring complexity (with $p=1$ representing the most complex wiring network)

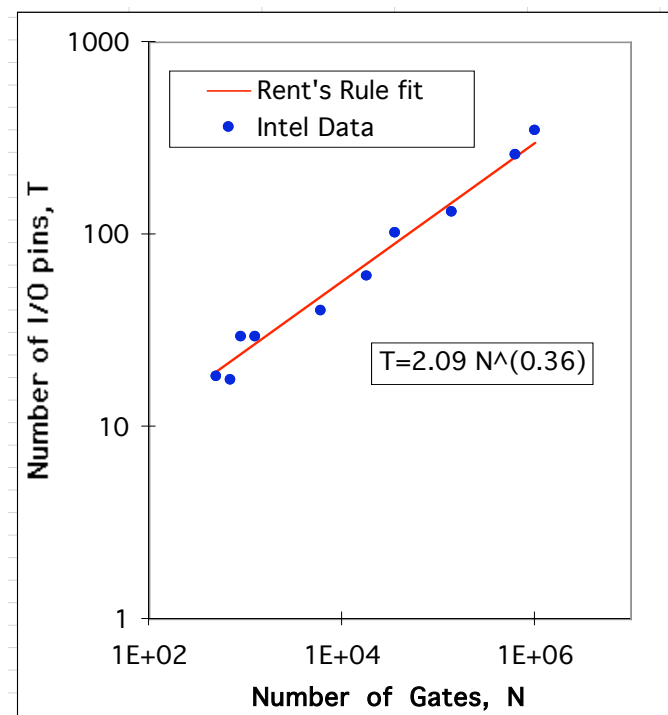


Fig. 12

The wire length distribution can be determined through the recursive application of Rent's Rule throughout an entire logic system. To determine all the shortest wires in a logic system, the recursive property of Rent's Rule is used, where the logic system is

divided into logic gates and Rent's Rule is applied to the interconnects between closest neighbor gates. This determines the number of interconnections between the closest logic gates. The longer wires are similarly determined by clustering the logic gates in growing numbers until the longest interconnects are found.

The wire length distribution can be described by $i(l)$, an Interconnect Density Function (i.d.f.), or by $I(l)$, the Cumulative Interconnect Distribution Function (c.i.d.f.) which gives the total number of interconnects that have length less than or equal to l (measured in gate pitches), and is defined as,

$$I(l) = \int_1^l i(x) dx \quad (13)$$

where x is a variable of integration representing length and l is the length of the interconnect in gate pitches. The derivation of the wire-length distribution in an IC is based on Rent's Rule. To derive the wire-length distribution, $I(l)$ of an integrated circuit, the latter is divided up into N logic gates, where N is related to the total number of transistors, N_t , in an integrated circuit by $N = N_t / \square$, where $\square$ is a function of the average fan-in ($f.i.$) and fan-out ($f.o.$) in the system. The gate pitch is defined as the average separation between the logic gates and is equal to $\sqrt{A_c / N}$ where A_c is the area of the chip.

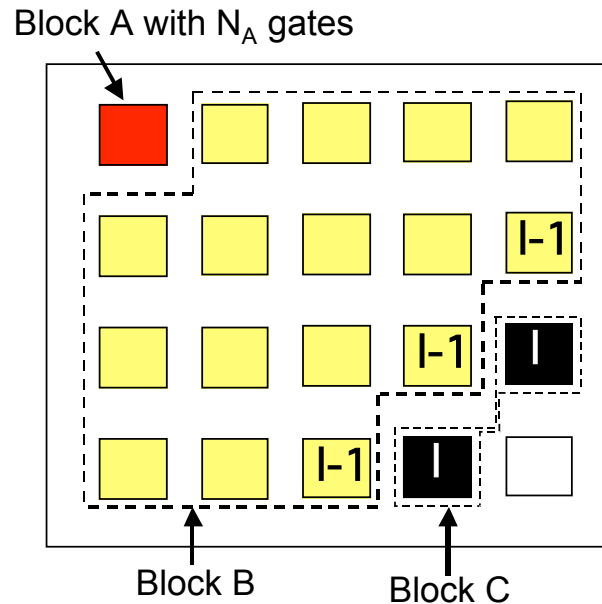


Fig. 13 Schematic view of logic blocks used for determining wire length distribution.

In order to derive the complete wire length distribution for a chip, the stochastic wire length distribution of a single gate must be calculated. The number of connections from the single logic gate in Block A to all other gates that are located at a distance of l gate pitches is determined using Rent's Rule. The gates shown in Fig. 13 are grouped into three distinct but adjacent blocks (A, B, and C), such that a closed single path can encircle one, two, or three of these blocks. The number of connections between Block A and Block C is calculated by conserving all I/O terminals for blocks, A, B, and C,

which states that terminals for blocks A, B, and C are either inter-block connections or external system connections.

Hence, applying the principle of conservation of I/O pins to this system of three logic blocks shown in Fig. 13 gives,

$$T_A + T_B + T_C = T_{A-to-C} + T_{A-to-B} + T_{B-to-C} + T_{ABC} \quad (14)$$

where T_A , T_B , and T_C are the number of I/Os for blocks A, B, and C respectively. T_{A-to-C} , T_{A-to-B} , and T_{B-to-C} are the numbers of I/Os between blocks A and C, blocks A and B, and between blocks B and C respectively. T_{ABC} represents the number of I/Os for the entire system comprising of all the three blocks. From conservation of I/Os, the number of I/Os between adjacent blocks A and B, and between adjacent blocks B and C can be expressed as,

$$T_{A-to-B} = T_A + T_B - T_{AB} \quad (15)$$

$$T_{B-to-C} = T_B + T_C - T_{BC} \quad (16)$$

Substituting (15) and (16) in (14) gives,

$$T_{A-to-C} = T_{AB} + T_{BC} - T_B - T_{ABC} \quad (17)$$

Now the number of I/O pins for any single block or a group of blocks can be calculated using Rent's Rule. If we assume that N_A , N_B , and N_C are the number of gates in blocks A, B, and C respectively, then it follows that,

$$T_B = k(N_B)^p \quad (18)$$

$$T_{AB} = k(N_A + N_B)^p \quad (19)$$

$$T_{BC} = k(N_B + N_C)^p \quad (20)$$

$$T_{ABC} = k(N_A + N_B + N_C)^p \quad (21)$$

where $N = N_A + N_B + N_C$. Substituting (18)-(21) in (17) gives,

$$T_{A-to-C} = k \left[(N_A + N_B)^p - (N_B)^p + (N_B + N_C)^p - (N_A + N_B + N_C)^p \right] \quad (22)$$

The number of interconnects between Block A and Block C (I_{A-to-C}) is determined using the relation,

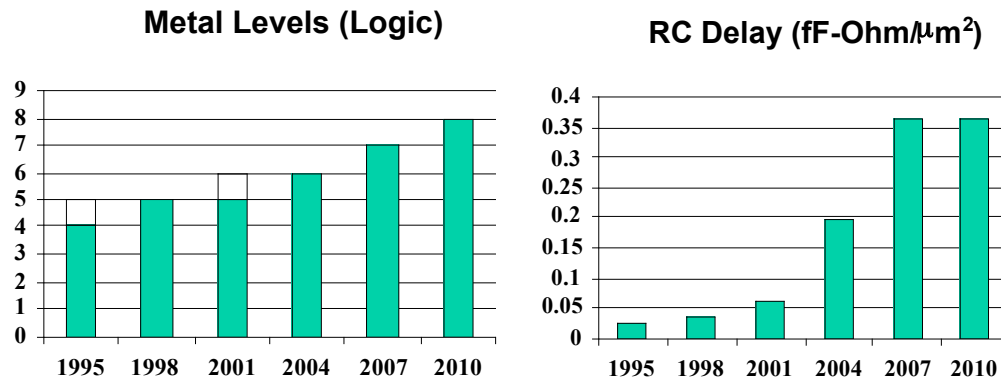
$$I_{A-to-C} = \alpha k (T_{A-to-C}) \quad (23)$$

Here α is related to the average fan-out (f.o.) by,

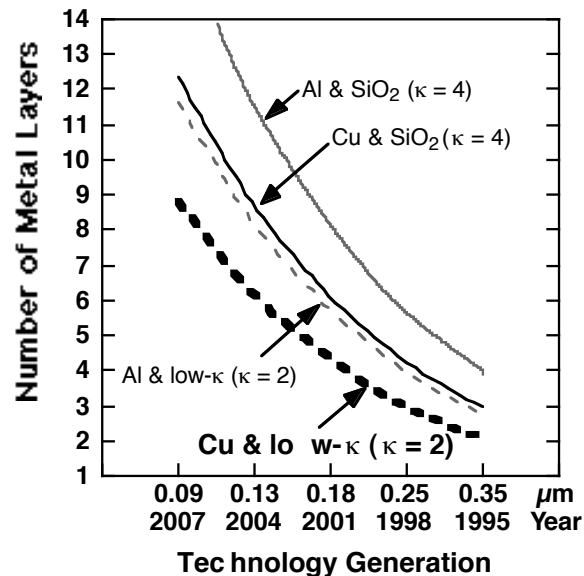
$$\alpha = \frac{f.o.}{1 + f.o.} \quad (24)$$

Equation (23) can be used to calculate the number of interconnects for each length l in the range from one gate pitch to $2\sqrt{N}$ gate pitches, to generate the complete stochastic wire-length distribution for the logic gate in Block A.

Interconnect Technology Roadmap

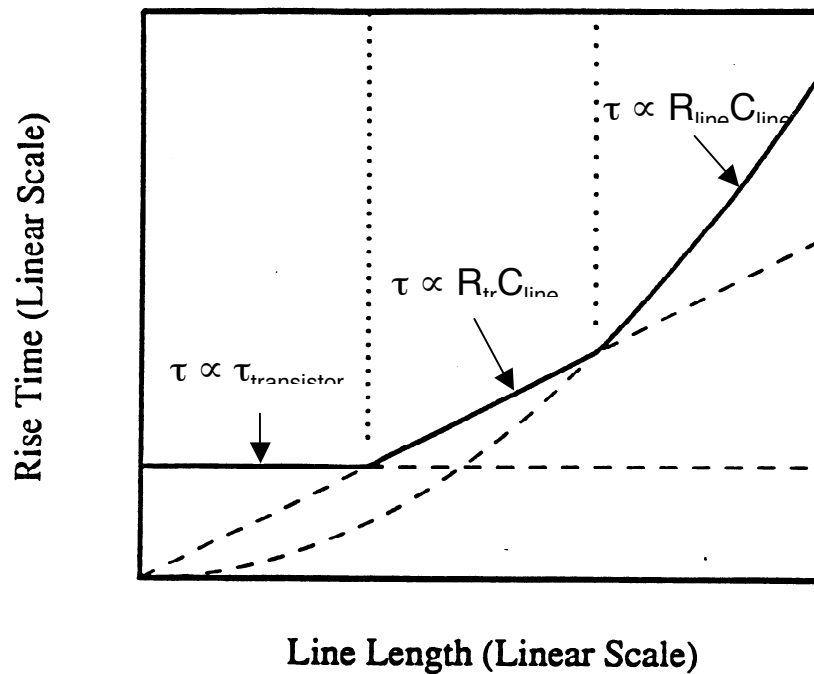
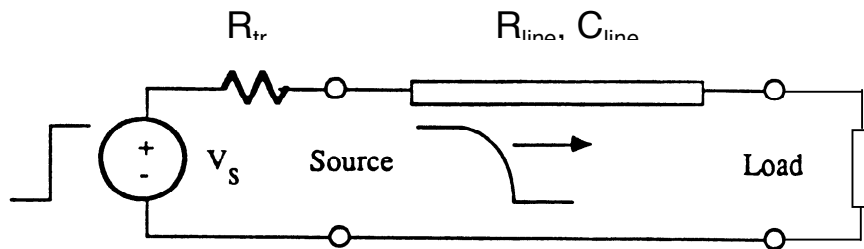


Number of metal layers for various interconnect materials options



- By using lower resistivity material wire resistance can be minimized
- By using lower K dielectric capacitance can be minimized
- An interconnect made with Al and SiO₂ placed at the global level can be placed at the semiglobal level if made made with Cu and a low-K material
- Interconnect density is larger at the semiglobal level and thus smaller number of levels are needed.

Circuit Speed Dependence on Transistor and Interconnect



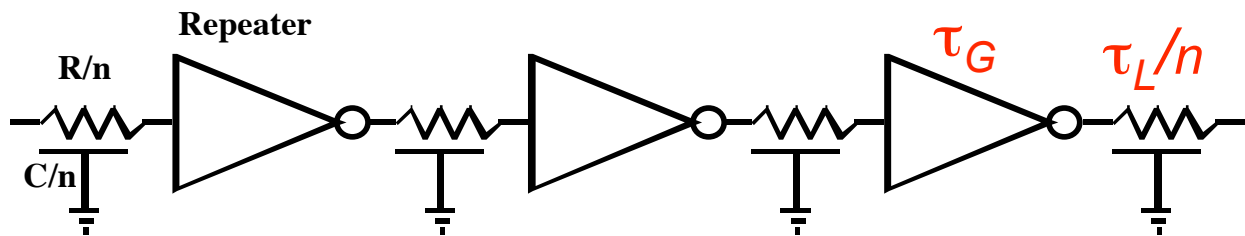
1. Small line length - Transistor speed governs the circuit speed.
2. Medium line length - Transistor resistance and line capacitance govern the circuit speed
3. Long line length - Line resistance and line capacitance govern the circuit speed

Note that inductance has been ignored here. At higher frequencies it will become very important and would not be neglected.

Reduce resistivity of the conductor and dielectric constant of the dielectric material

Repeaters

A first option for reducing RC delays is to use better interconnect materials when they are available and appropriate. However, for very long wires the delay can be substantially larger than the gate delay. For instance, in a 0.25 μm technology the gate delay is about 25 psec, however, a 5 mm long aluminum wire with 0.25 μm x 0.25 μm cross section has a delay of 1 nsec. It is possible to reduce the propagation delay by introducing intermediate buffers, generally known as *repeaters*, in the interconnect line.



From Eq. 6 the delay of a line is

$$\tau_L = \frac{3.56 \cdot K_{ox} \epsilon_o \rho}{\lambda^2} L^2 \quad (6)$$

By breaking a long interconnect line into n smaller lines the propagation delay of each line is reduced quadratically

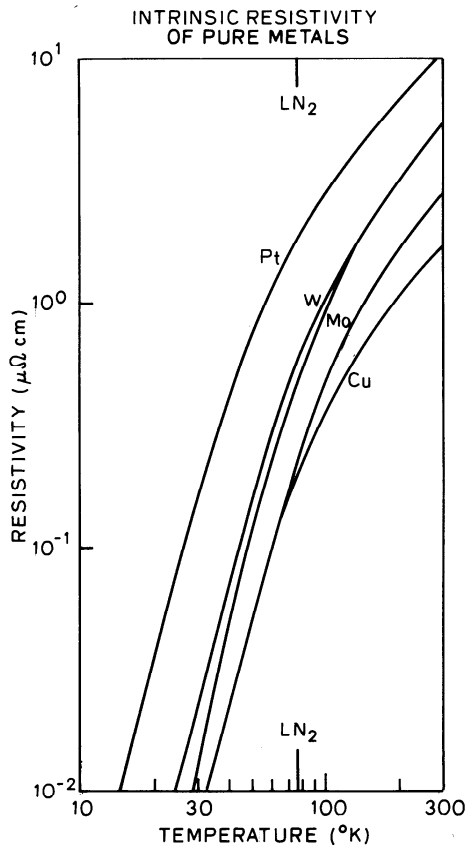
$$\tau_{L/n} = \frac{3.56 \cdot K_{ox} \epsilon_o \rho}{\lambda^2} \left(\frac{L}{n} \right)^2 \quad (9)$$

The total wire delay is thus

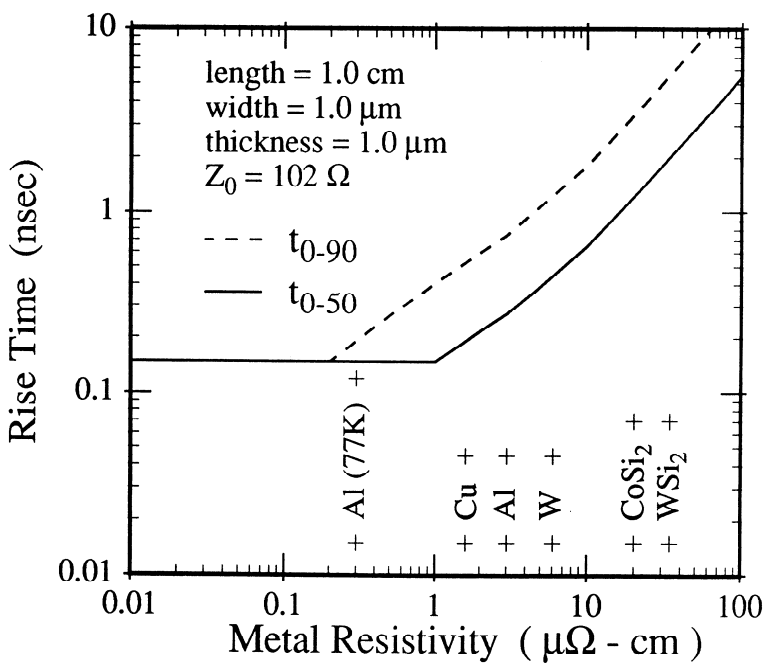
$$(\tau_{L/n} + \tau_G)n = \frac{3.56 \cdot K_{ox} \epsilon_o \rho}{\lambda^2} \left(\frac{L^2}{n} \right) + n\tau_G \quad (10)$$

As long as the gate delay is small the total wire delay is reduced substantially. This gain results at the cost of increased chip area occupied and extra power consumed by the repeaters.

Effect of lowering temperature



Cooling from room temperature to 77K reduces the resistivity by about an order of magnitude.



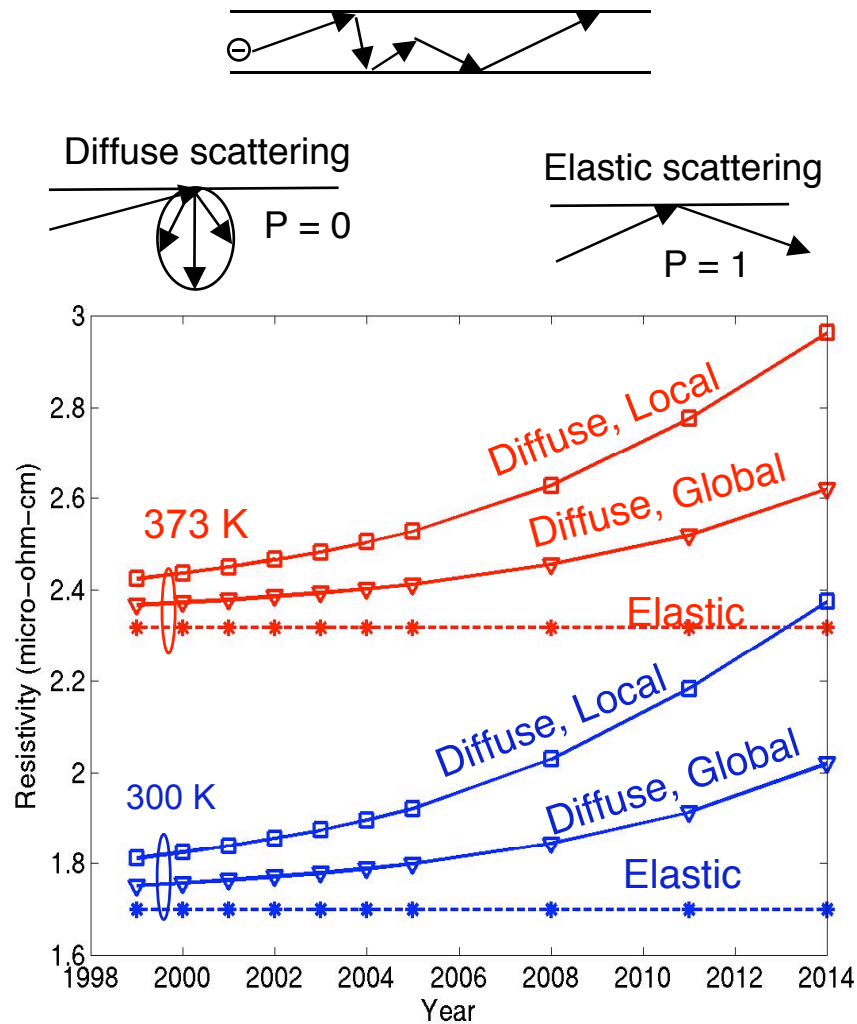
These simulations were done for an impedance matched line. At low values of line resistance the transistor impedance dominates the delay.

Ref:
Schroeder, Nishi & Saraswat

Will superconductors really improve the circuit speed?
Cooling conventional conductors to 77°K may be sufficient.

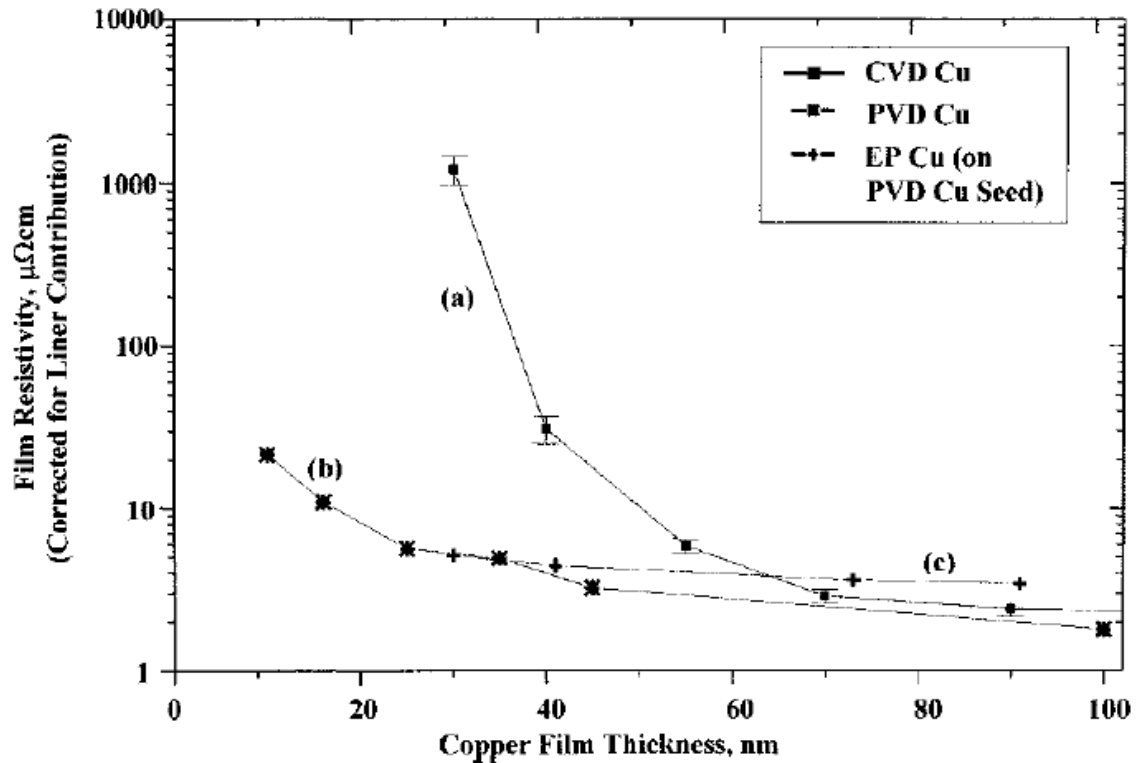
Thin Film Effects

- Effect of Electron Scattering $\Rightarrow$ Reduced mobility as dimensions decrease



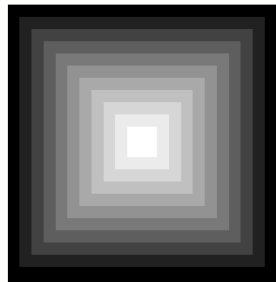
Resistivity of Cu wires for dimensions recommended by ITRS'99

- Surface electron scattering increases resistivity.
- Most metals today exhibit $P \approx 0.5$
- Real chips operate at higher temperatures
- With scaling the wire cross section is decreasing. As a result the surface scattering is becoming more serious.

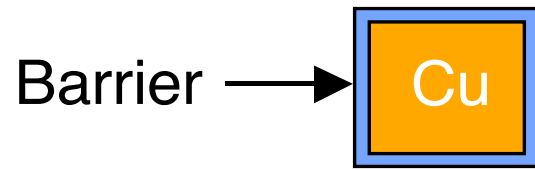


(Ref: Meindl et al. Proc IEEE, March 2001)

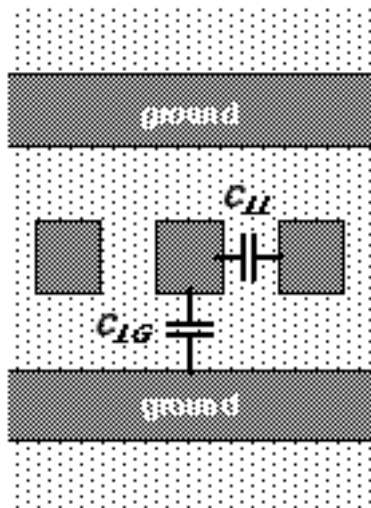
- Effect of Higher Frequencies - Carriers confined to outer skin increasing resistivity. These effects will become increasingly more important as the chips are operated in the GHz regime.



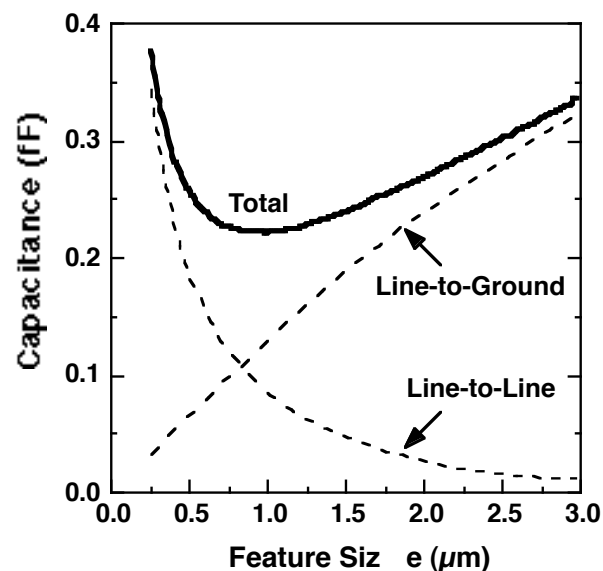
- Effect of Cu diffusion Barrier
 - Barriers have higher resistivity
 - Barriers can't be scaled below a minimum thickness



Crosstalk due to capacitive coupling



Higher Packing Density
↓
Decreased Space
Between Interconnects
↓
Higher crosstalk



Due to increased line-to-line capacitance there can be coupling of signal from one line to the adjacent line

Reduce dielectric constant of the dielectric material

Dielectric Materials

undoped plasma SiO₂
fluorinated SiO₂
spin-on glasses (silsesquioxanes)
organic polymers (e.g., polyimides)
fluorinated amorphous carbon (a-C:F)

Dielectric Constant

4.0-4.5
3.5
2.2-3.0
2.0-3.9
2.1-2.3

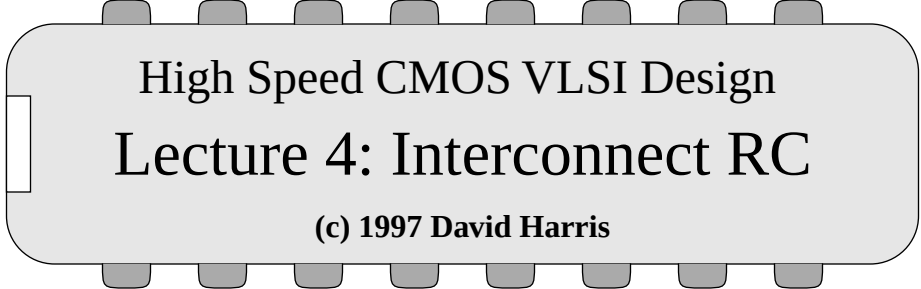
nanoporous dielectrics (e.g., xerogels)	1.2-2.2
Air	1

Requirements of the interconnection materials

- Low resistivity of conductors
- Low capacitance => low dielectric constant
 - Low RC delay
 - Low cross talk
 - Low power dissipation (CV^2f loss)
- Resistance to electromigration
- Ease of deposition of thin films of the material
- Ability to withstand the chemicals and high temperatures required in the fabrication process
- Ability to be thermally oxidized
- Good adhesion to other layers - low physical stress
- Stability of electrical contacts to other layers
- Ability to contact shallow junctions and provide low resistance
- Good MOS properties
- Ability to be defined into fine patterns - dry etching

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High Speed CMOS VLSI Design

Lecture 4: Interconnect RC

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1.0 Introduction

Once upon a time, the delay of a circuit was well approximated as the sum of the delays of the gates. Wires could be viewed as equipotential nodes that instantly carried the voltage to all ends. As transistors have shrunk, the relative delay of gates has decreased. At the same time, the delay of wires per unit length has actually increased. Thus, wires now are a very important part of total delay.

Wire delay comes from two sources. One is the intrinsic, speed-of-light delay. The second is the lossy nature of on-chip wires; because the resistance of such wires is very high, the wires form RC circuits. Speed of light delay is proportional to the length of the wire, while RC delay increases with the square of the wire length. Thus, for long wires, RC delay dominates. For shorter wires, speed of light delay still dominates. However, such delay over a short wire is still relatively small compared to gate delay.

In addition to increasing circuit delay, wires are subject to coupling noise. According to Maxwell's equations, changes in voltages and currents create electric and magnetic fields which in turn cause changes in voltages and currents on nearby conductors. These effects are called capacitive and inductive crosstalk, and get worse as conductors move closer together and edge rates get sharper. They now can greatly exceed the noise margins of many circuits unless interconnect is carefully designed.

We will first examine the RC wire effects which are most important in current circuits. We'll see how to compute wire resistance and estimate wire capacitance, then use these two properties in wire delay models. We'll also look at capacitive crosstalk and circuit failures caused by IR drops, electromigration, and self-heating. Since delay and crosstalk are so severe, we will explore how to optimize wire geometries, add repeaters, and drive buses to improve performance.

In the next chapter, we'll consider how RC effects scale with decreasing feature size. Then we'll address inductance, an effect which becomes more significant for wire pitches in the one micron range. We'll look at inductive delay effects, di/dt noise, and inductive crosstalk and mutter some banal warnings since little concrete is known about the subject.

2.0 Wire Resistance

The resistance of a wire can easily be computed from its resistivity ρ , length L , width W , and thickness t :

$$R = \rho L / Wt \quad (\text{EQ 1})$$

Since the thickness is not under designer control, resistance is frequently rewritten as $R_{\square} L / W$, where $R_{\square}$ is the sheet resistance having units of ohms / square. Typical values for various interconnect levels are:

TABLE 1. Typical Interconnect Sheet Resistances

Material	$R_{\square}$
diffusion	10
polysilicon	10
thin metal (m1, m2, ...)	0.07
thick metal (top level)	0.03

Without special treatment, diffusion and polysilicon would have much higher resistances that might degrade performance. Therefore, they are usually covered by a thin layer of a refractory metal such as tungsten which alloys with the silicon to form a “silicide” and reduce resistance to approximately 10 ohms/square. Even so, polysilicon should only be used for short interconnect within cells because it is still rather resistive. Diffusion should never be used for routing in high-performance circuits because it has tremendous parasitic capacitance as well as resistance.

Lower metal layers are usually thin, allowing very tight spacing but higher sheet resistance. Metal 1 can be especially thin and resistive. Higher metal layers intended for power distribution and critical signals are thicker and built on a wider pitch, reducing sheet resistance. Most wires are aluminum, but extensive research on other materials with slightly lower resistivity such as copper and gold are beginning to pay off in commercial processes. Such materials can have nasty effects if they contaminate transistors, so exotic schemes are used to prevent their infiltration of the substrate.

Every time a signal crosses metal layers, it must pass through a contact cut, adding further resistance. Such cuts generally have 1-10 ohms of resistance. Low-resistance connections therefore use an array of many cuts.

3.0 Wire Capacitance

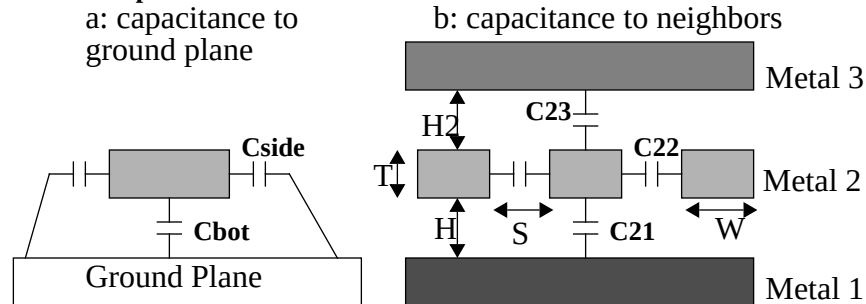
Wires also have capacitance to the substrate and to other nearby wires. A handy rule of thumb is that most wires have about 0.2 fF/ μm of total capacitance, about 1/10 that of a transistor gate. Therefore, when estimating sizes, one can treat 10 λ of interconnect as about 1 λ of gate loading.

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In the good old days, wires were much wider than thick. They could therefore be approximated as having parallel plate capacitance to the grounded substrate. Now, wires often have comparable thickness and width and are closer to adjacent wires than to layers above or below. Moreover, the layers above and below are much closer than the substrate. Thus, a negligible part of a wire's capacitance is to ground; instead it is to nearby wires which may transition and couple noise into the hapless wire.

These capacitances are shown in Figure 1:

FIGURE 1. Wire Capacitances for Metal 2



If all the capacitance is from a conductor to a solid ground plane, it can be represented as the sum of a parallel plate capacitance on the bottom and fringing capacitance on the sides. There are many physical and empirical models for such computing such capacitances relatively accurately¹. Unfortunately, this situation is rare in actual designs; instead, coupling occurs to neighbors on both sides and on the top and bottom. No closed-form equations are possible in general; instead, capacitances must be extracted with a 2 or 3-dimensional field solver using Poisson's equation. Many CAD tools avoid this time-consuming process by pre-extracting tables of capacitances for various geometries and interpolating between the tabulated results closest to the actual geometries of the circuit. Such a table for a 0.8 micron 4-level process can be found at:

<http://www-leland.stanford.edu/class/ee371/lib/wirecap>

Even in such a process, over 75% of the capacitance of a metal3 line over substrate can be to adjacent lines because the adjacent lines are much closer than the substrate. In more advanced processes, the coupling to adjacent lines gets even larger as the lines get closer. In a later section will analyze the amount of crosstalk such coupling can cause.

4.0 Wire Delay Modeling

Wires contribute delay from both the speed-of-light limit and the RC delay. For long wires, the RC term dominates because it increases quadratically with length while speed-of-light delay increases linearly. For short wires, speed-of-light would matter, except that

1. See E. Barke, "Line-to-Ground Capacitance Calculation for VLSI: A Comparison," IEEE Transactions on Computer-Aided Design, Vol. 7, No. 2, Feb. 1998, pp. 295-298.

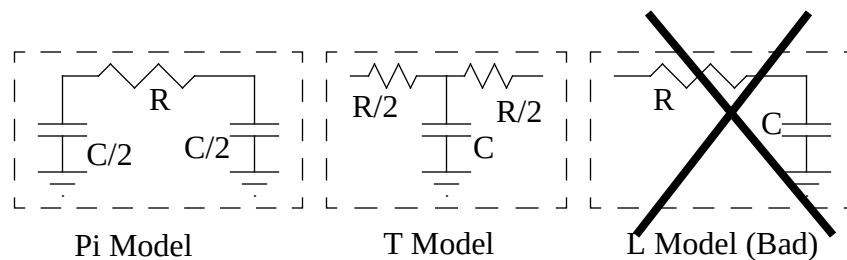
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the wires are short enough that the delay is usually assumed to be negligible. This assumption will change in the future as picoseconds of delay cease to be negligible.

4.1 Models

The distributed RC delay can be modeled by breaking up a wire into one or more segments and using a lumped model for each segment. If each segment has resistance R and capacitance C , popular lumped models include:

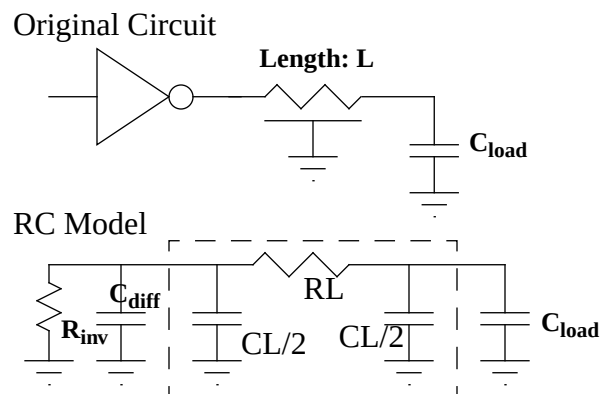
FIGURE 2. Lumped models of wires



Note that R and C depend on the length of the segment. The Pi model and T model, named for their shapes, are both good models. The Pi model can be slightly more convenient because it has fewer circuit nodes. The L model is much less accurate, requiring many more segments for the same accuracy of delay estimation. For example, the delay of a line can be modeled to within 5% accuracy with 4 Pi segments or 64 L segments! Thus, it is usually reasonable to model long wires with 4 Pi segments in simulation and with a single Pi segment for hand calculation.

The delay of the driver must also be included when estimating the delay of a wire. For example, we can compute the delay of an inverter driving an L mm wire with resistance R/mm and capacitance C/mm , inverter drive resistance R_{inv} , drain capacitance C_{diff} , and total load on the wire C_{load} :

FIGURE 3. Delay Estimation Example



The Elmore delay of the model is:

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$$R_{inv} * (C_{diff} + CL/2) + (R_{inv} + RL) * (CL/2 + C_{load}) \quad (EQ 2)$$

This can be rewritten as:

$$R_{inv} * (C_{diff} + C_{load} + CL) + RCL^2/2 + RLC_{load} \quad (EQ 3)$$

This expression has three terms. The first represents the inverter driving it's own diffusion, the load capacitance, and the lumped capacitance of the wire. The second is the quadratic delay of the wire's self-loading. It is proportional to $RC/2$, rather than RC , because the capacitance is distributed along the wire rather than lumped at the end. The third term is the extra delay contributed by the wire resistance discharging the load capacitance.

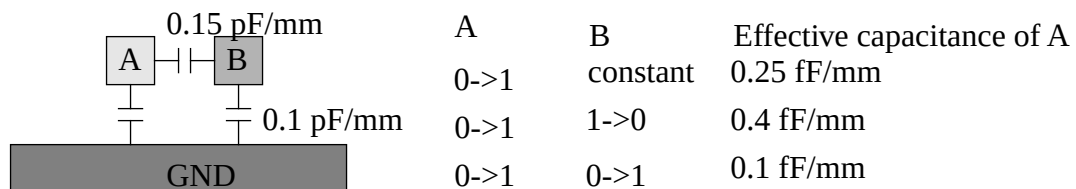
Remember that the units in this equation are somewhat curious. R_{inv} is Ω . R is $\Omega/\mu m$. C_{diff} and C_{load} are fF, while C is fF/ μm .

5.0 Capacitive Crosstalk

The RC models presented above assume that all capacitance is to ground, or some other node which is not transitioning. When the capacitance couples to another wire which might be switching, the crosstalk between wires causes noise and change in delay.

When two adjacent wires switch opposite directions, their delay is longer than if one switched while the other remained steady. This is because the capacitance between the wires sees a total of twice as much voltage change and thus requires twice as much current to be charged up. We can model this case for delay with a capacitor to ground twice as big as the capacitance between lines. For example, if a wire had 0.1 pF/mm capacitance to ground and 0.15 pF/mm capacitance/mm to an adjacent wire which might switch in the opposite direction, it can be modeled as a total of 0.4 pF/mm of capacitance to ground for delay purposes. Conversely, if the adjacent wire is switching in the same direction, the coupling capacitor will see zero voltage change and have no effect on delay. In the previous example, it could be modeled as 0.1 pF/mm of total capacitance to ground. At first, this seems good because the delay will be less. However, the delay of the wire becomes much less predictable because its effective capacitance changes by a factor of 4 depending on what the neighboring line does. If delays of two signals must be matched for some reason, the variation in delay caused by such coupling is very important.

FIGURE 4. Delay effects of crosstalk



Worse yet, the capacitive crosstalk introduces noise that could corrupt a signal. We can conservatively model crosstalk noise with a capacitive voltage divider. Suppose victim

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signal A is supposed to be 0. Adjacent aggressor signal B rises. The coupling between B and A tends to cause A to rise as well. The amount A rises depends on the ratio of coupling capacitance C_{AB} to total capacitance $C_{A-GND} + C_{AB}$. It is also influenced by the driver of A, which will fight the crosstalk, trying to maintain A at a 0. If the crosstalk happens fast enough that the capacitive impedances are much lower than the driver resistance, the driver will not help much and the voltage noise on A is:

$$C_{AB} / (C_{AB} + C_{A-GND}) * V_{DD} \quad (\text{EQ 4})$$

When $C_{AB} = C_{A-GND}$, the noise will be 50% and could cause a false transition on a gate at the end of the wire. Most static circuits can tolerate such noise glitches. Dynamic circuits, however, will be corrupted irrecoverably. Therefore, dynamic nodes are especially sensitive to coupling. Moreover, in a bus, a wire can receive coupling from both sides. In fact, it is even possible to receive coupling from orthogonal wires on layers above and below. Usually, this coupling is negligible. However, if a wide bus above or below switches entirely in one direction, the coupling can be significant. Most of the time, this coupling is ignored by verification tools. One day, chips may start to fail in a data-dependent way from such coupling on other layers!

When a node is actively driven, the small driver resistance does reduce the amount of coupling noise. Simulation may be the only way to determine exactly how much the resistance helps.

6.0 Other Wire Effects

In addition to delay and coupling noise, wires are susceptible to IR drops, electromigration, and self-heating.

6.1 IR Drops

Wires such as power busses which carry large DC currents droop in voltage due to the resistance of the wire multiplied by the current carried on the wire. This is bad because, for instance, it reduces the value of V_{DD} seen at the center of a chip when the power supply is driven by pads around the perimeter.

IR drops get to be a bigger problem as lower power supplies are used. For a fixed wattage, a lower V_{DD} means a higher I_{DD} flows through the power distribution network. For the same power network resistance, a higher IR drop is caused. Moreover, this drop is even a larger fraction of V_{DD} !

For example, a 100 watt chip operating at 2V requires 50 amps of supply current! If IR drop should be no more than 5% of V_{DD} , this requires a power network with only 2 m Ω of resistance. To meet the same spec, a 1V chip requires 100 amps of current and can only tolerate 0.5 m Ω of resistance in the power distribution network! Remembering that a typical top-level metal layer has a resistance of about 30 m Ω / square, the power network can only have a small fraction of a square of size. Even a solid power plane may be insuffi-

cient; instead power may be distributed in a thick plane in the package and dropped all over the chip with C4 bumps to reduce the resistance.

6.2 Electromigration

When high currents flow through a wire, they create an “electron wind” which can apply force on metal atoms in the wire. The atoms may actually move slightly, in an effect known as electromigration. If electromigration continues for a long time, it may actually cause a break in a wire, resulting in an open circuit and chip failure. This is especially unpleasant because it may happen after years of operation.

Such failure is avoided by limiting the current flowing through a wire. DC current is worst because it pushes atoms in a single direction. AC currents are less of a problem for electromigration because they push the atoms back and forth without a net average movement. Thus, power and ground lines are most prone to electromigration failure. A rule of thumb for many processes is to allow no more than 1 mA of current per micron of wire width. Wider wires may tolerate somewhat higher current densities than this. Processes with very thin wires tolerate less current. Check with your vendor or be conservative with wire widths.

6.3 Self-heating

IR drops in wires cause the wires to heat up. Oxide is a good insulator, so the heat may build up in the wires, resulting in the wires being hotter than the chip. This can increase wire resistance and degrade performance. In extreme cases, it is possible to experience thermal runaway, in which wires get hotter, thus become more resistive, causing larger IR drops and more heat until the wire melts down!

Such self-heating sets minimum wire widths even for lines carrying only AC signals.

7.0 Optimizing Wire Geometry

We have seen that wire resistance and capacitance cause significant amounts of delay and noise. Thus, optimizing wires can be as important as optimizing gates.

The best way to optimize wires is to keep them as short as possible. This requires careful floorplanning and good exploitation of datapath structure. Nevertheless, it is inevitable that a chip will have some long wires. What can we do to minimize delay and noise of these wires?

Wire design is fairly simple because there are only a few degrees of freedom. The designer can control the width of each wire, the spacing between wires, and the arrangement of which wires are adjacent to which other wires.

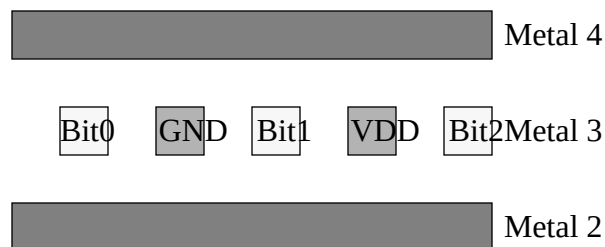
To reduce delay, the designer may adjust each of these factors. If the delay is dominated by the wire RC product, increasing wire width is often useful. This decreases wire resistance

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proportionally, while only slightly increasing total wire capacitance if the capacitance from the top and bottom of the wire is a small part of the whole. Increasing spacing between wires is also good because it decreases the total wire capacitance and the amount of coupling to adjacent wires. This is especially important when the adjacent wires might be switching and Miller-multiplying the coupling capacitance. Finally, interleaving busses such that the adjacent wires are not switching at the same time prevents Miller-multiplication of coupling capacitance. For example, a bus driven in phase 1 could be interleaved with a bus driven in phase 2 so no two adjacent wires switch at once.

The designer also may use these degrees of freedom to reduce crosstalk. The first tactic is to increase the spacing between lines, thus reducing coupling. At some point, this yields diminishing returns. The next option is to increase the width of the wire or to shield the wire with power/ground lines. Increasing the width of a wire is sometimes helpful because it increases the fraction of capacitance to the layers above and below rather than to an adjacent coupling wire. It also reduces the RC delay. In contrast, shielding the wire with a nearby minimum pitch power or ground line is extremely effective because it eliminates almost all capacitive coupling. It also increases the number of wires in the power and ground grids which may improve the power supply network. The drawback is that it may increase the total amount of capacitance on a wire and thus increase RC delay. Shielding every bit of a wide dynamic bus also has horrible effects on area.

FIGURE 5. Shielding adjacent lines of a dynamic bus



As mentioned earlier, the capacitance to adjacent wires and to the upper and lower layers is a complex function of wire width and spacing. It may be necessary to use a table of capacitances to understand how width and spacing influence noise and delay. The designer faces a constant tradeoff between delay and area; there is no single pitch optimum for all problems.